Literature review: Almogy & Yariv (LEOS 1994) and Vurgaftman et al. (IEEE JQE 1996)

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Almogy and Yariv's 1994 summary partly anticipates NL-02 and independently confirms it: our density-matrix solver reproduces their closed-form saturated susceptibility to five digits, and our ceiling condition gives their μ12/μ23 = √2. The uploaded Vurgaftman 1996 paper is not the one NL-02 cites (that is IEEE JQE 32, 1334, August 1996), but it still opens three threads. The review adds eight research threads and four items to read before NL-02 is submitted.

What each paper does

Paper	What it does	Key result	Relevance to us
G. Almogy, A. Yariv, "Quantum well intersubband nonlinear optics: beyond perturbation theory," Proc. LEOS'94, vol. 1, pp. 179–180 (invited NO3.1)	Rotating-wave, non-perturbative solution of the optical Bloch equations for a dc-biased two-level well (electro-optic effect) and a three-level ladder driven by an intense pump (SHG)	EO: the Stark shift moves the Lorentzian exactly, so the EO coefficient saturates and changes sign and the index change is capped by the transition's own index. SHG: $\chi_{eff} = \chi_{res} / [1 + 2\Omega^2 T_2^2 (1 + 3T_1/T_2)]$, $\Omega = \mu E/2\hbar$; SHG saturates at about half the absorption saturation intensity; resonance is optimal below saturation and detuning above it "if propagation length limitations are removed"; SHG vanishes asymptotically for $\mu_{12}/\mu_{23} = \sqrt{2}$. Efficiency plotted for one absorption length, peaking near 1% with I_SH saturating at I_sat/16	High: closest prior art to NL-02 Secs. 5–6; an exact benchmark for our solvers; points to strong-field SHG in symmetric wells (Levinson et al., PRA 1994) and to ISB birefringence (Almogy et al., APL 1993)

Paper	What it does	Key result	Relevance to us
I. Vurgaftman, P. N. Freeman, P. K. Bhattacharya, J. Singh, "Time-domain optical wave propagation in disordered and nonuniform guiding structures," IEEE JQE 32, 2095–2104 (Dec 1996)	Implicit, locally one-dimensional finite-difference solver of the 2-D scalar wave equation for pulses in waveguides with roughness and longitudinal nonuniformity	Etched-mirror reflectivity falls as $\exp(-a \cdot L \cdot \delta/\lambda^2)$, $a \approx 10$, for roughness depth δ and correlation length L; a lateral waveguide couples light into an absorbing HBT collector within about 10 μm	Low for NL-02 directly. Useful for: background loss from sidewall roughness (which sets NL-02′s high-power ceiling), segment boundaries in electrode-written QPM (N1), pulsed time-domain propagation, and waveguide-to-absorber coupling

Checks against our work

Every statement of Almogy and Yariv we could test agrees with NL-02 once their relaxation convention (each excited level decays straight to the ground state at 1/T1) is used. Script: `studies/nl02/ay_check.py`.

Their statement	Our result	Verdict
$\chi_{eff}/\chi_{res} = 1/[1 + 2\Omega^2 T2^2(1 + 3T1/T2)]$	Full density matrix: identical to 5 digits for T1/T2 = 5–50 and ΩT2 = 0.03–3. Rate equations: $1/(1 + 6\Omega^2 T1T2)$, i.e. the same without the coherent term $2\Omega^2 T2^2$	Exact agreement; explains why our rate equations are accurate at resonance (error below T2/3T1, about 2% for T1/T2 = 15)
SHG vanishes asymptotically for $\mu12/\mu23 = \sqrt{2}$	Our closed-form ceiling condition r23 = (t32+t31)/(1+t31) gives $\sqrt{2}$ for decay to ground; density matrix: $\chi \cdot I$ falls as $1/I$ there, constant for equal dipoles	Exact agreement

Their statement	Our result	Verdict
Detuning preferable above saturation if propagation length is not limited	Quantitative rule (detune pump by γ√(I/Is), harmonic oppositely to keep α2ω = 2αω) recovers Theorem 1; 15–31× in the ladder, 3–6× in qwkit structures	Qualitative prior art; NL-02 now credits it
Efficiency for one absorption length peaks near 1%, I_SH saturates (I_sat/16)	With the length optimized there is no ceiling (Sec. 5); a fixed length behaves like our thin film	Consistent; their ceiling is a fixed-length effect
EO: index change capped by the transition's own index	Our identity analogue for Stark phase modulators: phase per neper of loss ⩽ Q/2γ, exactly, independent of doping, dipole and length (Q = Stark shift)	New bound, consistent with their cap and with Almogy's thesis preferring absorption modulation

What changed in NL-02

- Introduction and Discussion now credit Almogy and Yariv's non-perturbative treatment, their fixed-length efficiency, their qualitative detuning remark, and their limits papers (J. Nonlinear Opt. Phys. Mater. 4, 401, 1995; Caltech thesis, 1995); also Khurgin (JOSA B 1989), Almogy, Segev and Yariv on index-induced phase matching (Opt. Lett. 1994) and Goldoni's χ(2) quantum limit (J. Appl. Phys. 2001).
- Sec. 2 gives the analytic reason the rate equations work at resonance: the exact solution differs only by the coherent term $2\Omega^2T2^2$, a relative error below T2/3T1. Our solver reproduces the exact form to five digits.
- Sec. 5 notes that our ceiling condition reproduces their √2.
- Sec. 6 opens by crediting their detuning remark and states that Theorem 1 makes it quantitative.
- The novelty that remains: the identity $|\chi(2)|^2 = Q\cdot\alpha\omega\cdot\alpha2\omega$; Theorem 1 at the optimum length; Theorem 2 for any saturation, length and QPM; the no-ceiling result with the length optimized and its closed-form slope; the υ = 1 detuning rule; the thin-film contrast; and the qwkit validation. Whether the 1995 review or thesis already contains the identity or the no-ceiling result is the main open risk.

New research threads

ID	Thread	Origin	Expt	Priority
NL-08 (upgraded)	Universal phase-per-loss bound for ISB Stark phase modulators: $\Delta\varphi \leqslant (Q/2\gamma)\cdot\alpha L$, so a π shift costs $\geqslant 2\pi\gamma/Q$ nepers (13.6 dB at $Q = 2\gamma$); add bandwidth and RC limits	Almogy–Yariv EO result + NL-02 identity; verified numerically	No	P1
NL-18 (broadened)	Gain–nonlinearity identity for intracavity DFG in QCLs: the same inverted transitions give gain and $\chi(2)$, so	$\chi(2)$	$^2 \propto$ gain product × (THz dipole)^2/γ^2, bounding THz conversion per mid-IR power	NL-02 identity applied to inversion
NL-22	Time-domain Maxwell–Bloch ISB propagation: sub-T1 pulses avoid population saturation, so pulsed SHG may exceed the steady-state limits; new qwkit module	Vurgaftman et al. time-domain method + NL-02 steady-state theorems	No	P1

ID	Thread	Origin	Expt	Priority
NL-23	Strong-field symmetry breaking in N1′s symmetric triple wells: optically written χ(2) sign gratings and even harmonics without bias	Levinson, Segev, Almogy, Yariv, PRA 49, R661 (1994)	Opt	P2
NL-24	Saturation-engineered ladders: choose z23/z12 and the relaxation paths (LO-phonon placement) to maximize the high-power slope s∞, or set z12/z23 = √2 for an SHG power limiter	NL-02 closed-form C∞ + Almogy–Yariv zero	No	P2
NL-25	Loss budget of resonant ISB ridge waveguides (sidewall roughness, free carriers, metal): background loss sets NL-02′s high-power ceiling	Vurgaftman et al. roughness model + NL-02	Opt	P2

ID	Thread	Origin	Expt	Priority
NL-26	Bound on sum-frequency upconversion detectors (mid-IR to near-IR) per absorbed pump, from the identity; waveguide-coupled absorber design	NL-02 identity + Vurgaftman HBT coupling example	No	P2
N1-b	Segment-boundary scattering in electrode-written QPM: confirm that sign reversal has no index step, unlike Stark on/off	Vurgaftman et al. nonuniform-waveguide method	No	Fold into N1

NL-17 (chirped or power-matched phase matching) also gains support: Almogy's thesis identified the ISB index as a possible dominant limit on conversion, and NL-02's density-matrix check finds it is the main correction under detuning.

Still to read before submission

- [ ] Vurgaftman, Meyer, Ram-Mohan, "Optimized second-harmonic generation in asymmetric double quantum wells," IEEE JQE 32(8), 1334–1346 (Aug 1996), doi 10.1109/3.511546 (the uploaded file is a different paper)
- [ ] Almogy, Yariv, "Resonantly-enhanced nonlinear optics of intersubband transitions," J. Nonlinear Opt. Phys. Mater. 4, 401–458 (1995), doi 10.1142/S0218863595000173: most likely place for prior forms of the identity or limits
- [ ] Almogy, Caltech PhD thesis (1995), doi 10.7907/3qj9-9777: contains "a limit on the second-harmonic intensity in a given propagation length" (scanned PDF)
- [ ] Almogy, Yariv, "Second-harmonic generation in absorptive media," Opt. Lett. 19, 1828 (1994)

Sources

- Almogy & Yariv, LEOS'94 (doi 10.1109/LEOS.1994.586950)
- Vurgaftman, Freeman, Bhattacharya, Singh, IEEE JQE 32(12), 2095–2104 (1996), uploaded copy (DOI not verified)
- Almogy thesis record, Caltech
- Levinson et al., PRA 49, R661 (1994)
- Almogy, Shakouri, Yariv, APL 63, 2720 (1993)
- Almogy & Yariv, JNOPM 4, 401 (1995)
- Goldoni, J. Appl. Phys. 89, 1755 (2001)
- Cominotti et al., PRB 107, 115431 (2023)
- Crossref record of the LEOS'94 paper